

Afterlife v1.2 — Easy & Exciting Spec

3 October 2026 · Status: awaiting review · Owner: Ujjwal (director) · Builder: Claude · Builds on: v1 (2026-10-03-afterlife-design.md) and v1.1 (2026-10-03-afterlife-v1.1-design.md), live at <https://afterlife-one.vercel.app>

1. Purpose and success criteria

A first-time player should start playing within seconds without reading anything, never get stuck or confused, and feel frequent rewarding moments. The calm spirit stays: no timers and no fail screens.

v1.2 is done when all of these are true:

- 1. A new player’s first tap on the board plants something (Moss is preselected), on any device.
- 2. One tap on a tile places the selected item, on touch as well as mouse.
- 3. After 3 s without a move, the best tile for the selected item glows.
- 4. Every level can always be finished. Running out of scrap gives a bonus crate instead of a dead end.
- 5. Big moves celebrate (combo bursts), progress milestones are felt (25/50/75%), and each win awards 1–3 stars, with the best saved.
- 6. Animations are about 30% faster than v1.1.
- 7. All existing tests still pass, the new logic has unit tests, and the Bus Stop and Rooftop solutions are re-proven.

2. Decisions

Topic	Decision
Placement	One tap places, on mouse and touch. The two-tap touch preview is removed. The desktop hover preview stays
Default selection	Moss (or the first plant type with seeds) is selected when a level starts. When the selected item runs out, the selection moves to the next available item: seeds in tray order, then scrap slot 0
Hint	After 3 s idle, <code>bestTile(state, selection)</code> glows using the highlight. The tutorial highlight takes priority. Any input resets the timer

Topic	Decision
No dead ends	A new engine move, bonus: allowed only when the tray and batches are empty and the level is not won. It puts ['crate'] in the tray and counts <code>bonusUsed += 1</code>
Gentler start	Bus Stop and Rooftop: target 0.6 → 0.5 and moss +2. Solutions are regenerated with the solver
Combo	Size = number of grew + spread events from one scrap placement. 4–6 “Nice!”, 7–10 “Lush!”, 11+ “Wild!”
Stars	3★: won, <code>bonusUsed = 0</code> , and leftover scrap ≥ 25% of the level’s total. 2★: won with <code>bonusUsed = 0</code> . 1★: won with a bonus
Milestones	Every upward crossing of 25%, 50% or 75% of progress triggers a flash ring, birds and a swell. Undoing below a mark and re-crossing it replays the moment, by design
Pacing	All animation timings in one <code>timing.ts</code> module, at about 0.7× v1.1

3. Rules detail

3.1 Bonus crate (engine)

- `GameState` gains `bonusUsed`: number (0 at start).
- Move gains { `type: 'bonus'` }. `applyMove` rejects it with 'bonus not available' unless the state is not won, the tray is empty and the batches are empty.
- On success: `tray = ['crate']`, `bonusUsed + 1`, and the event { `type: 'bonus'` } is emitted.
- **Stuck changes:** `isStuck` is true only when no tile is free (unblocked, with no object and no plant). The old “no scrap left” case becomes “bonus available” (`canBonus(state)`).
- **The controller** applies bonus automatically, right after a move leaves the tray and batches empty with the level not won. The HUD shows a short “Bonus crate!” sparkle on the tray.

3.2 Best-tile hint (pure)

`bestTile(state, selection)`: `Pos | null`. All candidate tiles are scanned in row-major order and ties are broken by first found.

- **Scrap:** among tiles where `placeScrap` is valid, the one maximising (new covered tiles after the move, then plant cells within radius).

- **Seed:** among tiles where placeSeed is valid, the one maximising (number of plant cells within Manhattan 2, then closeness to the board centre).
- **No selection, or no valid tile:** null.

3.3 Stars (pure)

starsFor(level, finalState) → 1 | 2 | 3

total = number of scrap items across all of the level's batches

left = tray length + items in the remaining batches

- **1★** if finalState.bonusUsed > 0.
- **3★** if left / total ≥ 0.25.
- **2★** otherwise.

Stars are computed when the won event fires. The save gains stars: Record<levelId, 1 | 2 | 3> and keeps the maximum.

3.4 Combo and milestones (pure)

- comboFor(events) → { size: number; label: 'Nice!' | 'Lush!' | 'Wild!' } | null. Only events that contain placedScrap count. Size is the number of grew + spread events. It returns null below 4.
- milestonesCrossed(prevProgress, nextProgress) → (25 | 50 | 75)[]: each threshold t with prev < t/100 ≤ next.

4. Presentation

- **Combo:** floating text rises 40 px from the scrap tile and fades over 900 ms (larger for “Wild!”). The particle burst count scales with size (8 + size). The combo sound cue combo plays a short arpeggio whose top note rises with the size bucket.
- **Milestone:** a light ring expands from the board centre (500 ms); 3–5 birds (a drawn V shape) fly across the sky in 2 s; a sound cue milestone plays. The meter gets tick marks at 25, 50 and 75%.
- **Stars:** on the win panel, three star icons fill in one by one (150 ms apart). The level cards show the best stars (☆/★).
- **Bonus:** a sparkle burst over the tray and a “Bonus crate!” toast for 1.5 s; sound cue bonus.
- **Reduce motion:** no floating text motion (the text shows and hides), no birds or rings, and instant star fill.

5. Pacing (src/render/scene/timing.ts)

	v1.1	v1.2
Grow	500	350
Sprout (+delay)	450 (+150)	320 (+100)

	v1.1	v1.2
Bloom open	600	420
Seed pop	250	180
Scrap drop	350	250
Ripple	400	280
Grow lag after drop	350	250
Harvest flight	150 + 450	110 + 310
Wash step	40	28
Camera turn step	450	350

6. Teaching updates

- **Tutorial texts** stay as they are, except step 2, which becomes “Tap a glowing tile to plant.” Moss is preselected, so the tutorial opens directly on step 2. Step 1 (“Tap Moss in your tray.”) only appears if the player deselects Moss before planting.
- **How to Play card 1** becomes “Tap a tile to plant. Your seed is already picked.”
- **Card 5** adds “Stuck? Wait a moment — the best tile glows.”

7. Testing

What	How
Bonus move, canBonus, the new isStuck	Engine unit tests; all 5 level solutions still pass
Gentler levels	Solver regenerates both levels; the solvability tests pass
bestTile, comboFor, starsFor, milestonesCrossed	Unit tests
Controller: one tap, default selection, auto-switch, auto-bonus	Unit tests
Save stars	Unit tests (default {}, max kept, old saves load)
HUD stars, ticks, bonus toast; App hint timer and star save	DOM tests
Visuals, pacing, phone and reduce motion	Headless Chrome checks; live check after release

8. Out of scope

New levels, leaderboards and accounts, daily challenges, a timer mode, and new plant types.